

Machine Learning - Lab 1 Regression Models

Group 13

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1. COMPARING BOTH MODELS USED

Linear Regression and Random Forest Regression were the models chosen for the completion of this lab. These two can provide different approaches as supervised learning for a regression problem. While the Linear Regression is a simple model that can be used as a reference despite its simplicity to find out if a simple solution is enough, the Random Forest Regression's tree combinations can detect nonlinear relations and more complex interactions among variables. In this way, it is clearly demonstrated if the higher complexity and execution time from the Random Forest contributes to a better solution than the Linear model.

Linear Regression

- Strengths: simple to implement, quick execution, easier interpretation, effective for linear relationships.
- Weaknesses: may not be enough for complex nonlinear relationships. The results of the measurements indicate that its performance is low for complex datasets.

Random Forest

- Strengths: captures nonlinear relationships, combines predictions from multiple decision trees, it surpasses in these three measurements to the LR for a complex dataset.
- Weaknesses: way more resource demanding to execute, harder interpretation than LR when used for a complex dataset, needing more attention to interpret scores.

As it is shown in the table taken directly from the code output, both before and after the tuning, the Random Forest Regression marks better scores than the Lineal one.

```
Linear Regression:
Metric  Before tuning  After tuning
   MAE    1725.530174   1725.530174
  RMSE    2516.131781   2516.131781
    R2       0.547590      0.547590

Random Forest Regression:
Metric  Before tuning  After tuning
   MAE     529.894957    544.445535
  RMSE    1220.524693   1227.614733
    R2       0.893547      0.892307
```

- **Mean Absolute Error:** being the measure for the average absolute difference between the predicted and the actual prices, Random Forest shows a lower difference, therefore Random Forest proves to be more accurate.

- **Root Mean Squared Error:** being the measure penalizing errors between prediction-value, has also a lower value in Random Forest, proving that it makes fewer large prediction errors.

- **R^2 / Coefficient of Determination:** being the measure of how much of the variation in the target variable is explained by the model. The Random Forest has higher value, which proves a higher accuracy.

For the dataset selected for this lab specifically, Random Forest Regression clearly outperforms LR checking the results, being substantially better for predicting the price due to the big amount of features influencing it.

2. MOST INFLUENTIAL FEATURES

Both models are isolated in pipelines and were tuned testing different configurations, so first the best configuration is taken and then divided in the preprocessor of the data and the model function.

```
best_rf_pipeline = cast(
    Pipeline,
    rf_search.best_estimator_
)
```

```
# preprocessing step
rf_preprocessor = best_rf_pipeline.named_steps[
    "preprocessor"
]

# trained Random Forest
rf_model = best_rf_pipeline.named_steps[
    "model"
]
```

Once separated the names of the transformed data in the preprocessor, we can take the name and the importance in the process, measured by the weight of each one in the decisions of the trees in the RF model. The higher the score is, the most influential the feature was for the decision.

```
feature_importance = pd.DataFrame({
    "Feature": feature_names,
    "Importance": importances
})

feature_importance = (
    feature_importance
    .sort_values(
        "Importance",
        ascending=False
    )
)
```

From there, the name of the features and the importance is represented on the next table. Despite the results, it is not possible to establish a direct causation because of the nature of the dataset, but a reduction in the impurity of the prediction through the decision trees.
The date of the flight (month of the year first, day of the month third) appears to be the most influential feature in the decision making, followed by flight duration.

